# Apex Distributed Systems LLC

ENTERPRISE CLOUD ARCHITECTURE AND DISTRIBUTED INFRASTRUCTURE

100 Montgomery Street, Suite 2400
San Francisco, CA, 94104
Tax ID: US-84-9921048 | Email: invoicing@apex.systems

# INVOICE

Invoice Number: INV-2026-0010
Issue Date: 2026-10-11
Due Date: 2026-11-10
Payment Terms: Net 30 Days
Currency: USD

**BILLED TO (CLIENT)**

**Cobalt Data Processing Ltd**

Platform and Cloud Infrastructure Engineering
Enterprise Technology Plaza, Floor 12
London, UK, 10001
Attention: Alistair Finch, VP Engineering
Email: billing@cobalt.io

**REMIT PAYMENT TO**

**JPMorgan Chase Bank, N.A.**

Account Holder: Apex Distributed Systems LLC
Account Number: **** **** 7821
Routing Number (ACH): 021000021
SWIFT / BIC: CHASUS33
Reference: INV-2026-0010

| # | SKU | DESCRIPTION | QTY | RATE (USD) | AMOUNT (USD) |
|---|---|---|---|---|---|
| 1 | ENG-INF-001 | Distributed consensus state-machine replica verification and Byzantine audit (Milestone 1) | 2 | $480.00 | **$960.00** |
| 2 | ENG-INF-002 | High-throughput reactive event ingestion pipeline tuning and partition testing (Milestone 2) | 3 | $350.00 | **$1050.00** |
| 3 | ENG-INF-003 | Native PostgreSQL read-replica connection pooling optimization via PgBouncer (Milestone 3) | 4 | $275.00 | **$1100.00** |
| 4 | ENG-INF-004 | Redis cluster cache stampede mitigation via probabilistic early expiration (Milestone 4) | 1 | $320.00 | **$320.00** |
| 5 | ENG-INF-005 | Zero-downtime schema migration verification for high-concurrency order ledger (Milestone 5) | 2 | $450.00 | **$900.00** |
| 6 | ENG-INF-006 | Virtual thread concurrency soak probe under simulated 10k connections (Milestone 6) | 3 | $390.00 | **$1170.00** |
| 7 | ENG-INF-007 | Distributed trace propagation audit and OpenTelemetry exporter alignment (Milestone 7) | 4 | $260.00 | **$1040.00** |
| 8 | ENG-INF-008 | Distributed consensus state-machine replica verification and Byzantine audit (Milestone 8) | 1 | $480.00 | **$480.00** |
| 9 | ENG-INF-009 | High-throughput reactive event ingestion pipeline tuning and partition testing (Milestone 9) | 2 | $350.00 | **$700.00** |
| 10 | ENG-INF-010 | Native PostgreSQL read-replica connection pooling optimization via PgBouncer (Milestone 10) | 3 | $275.00 | **$825.00** |
| 11 | ENG-INF-011 | Redis cluster cache stampede mitigation via probabilistic early expiration (Milestone 11) | 4 | $320.00 | **$1280.00** |
| 12 | ENG-INF-012 | Zero-downtime schema migration verification for high-concurrency order ledger (Milestone 12) | 1 | $450.00 | **$450.00** |


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| # | SKU | DESCRIPTION | QTY | RATE (USD) | AMOUNT (USD) |
|---|---|---|---|---|---|
| 13 | ENG-INF-013 | Virtual thread concurrency soak probe under simulated 10k connections (Milestone 13) | 2 | $390.00 | **$780.00** |
| 14 | ENG-INF-014 | Distributed trace propagation audit and OpenTelemetry exporter alignment (Milestone 14) | 3 | $260.00 | **$780.00** |
| 15 | ENG-INF-015 | Distributed consensus state-machine replica verification and Byzantine audit (Milestone 15) | 4 | $480.00 | **$1920.00** |
| 16 | ENG-INF-016 | High-throughput reactive event ingestion pipeline tuning and partition testing (Milestone 16) | 1 | $350.00 | **$350.00** |
| 17 | ENG-INF-017 | Native PostgreSQL read-replica connection pooling optimization via PgBouncer (Milestone 17) | 2 | $275.00 | **$550.00** |
| 18 | ENG-INF-018 | Redis cluster cache stampede mitigation via probabilistic early expiration (Milestone 18) | 3 | $320.00 | **$960.00** |
| 19 | ENG-INF-019 | Zero-downtime schema migration verification for high-concurrency order ledger (Milestone 19) | 4 | $450.00 | **$1800.00** |
| 20 | ENG-INF-020 | Virtual thread concurrency soak probe under simulated 10k connections (Milestone 20) | 1 | $390.00 | **$390.00** |
| 21 | ENG-INF-021 | Distributed trace propagation audit and OpenTelemetry exporter alignment (Milestone 21) | 2 | $260.00 | **$520.00** |
| 22 | ENG-INF-022 | Distributed consensus state-machine replica verification and Byzantine audit (Milestone 22) | 3 | $480.00 | **$1440.00** |
| 23 | ENG-INF-023 | High-throughput reactive event ingestion pipeline tuning and partition testing (Milestone 23) | 4 | $350.00 | **$1400.00** |
| 24 | ENG-INF-024 | Native PostgreSQL read-replica connection pooling optimization via PgBouncer (Milestone 24) | 1 | $275.00 | **$275.00** |
| 25 | ENG-INF-025 | Redis cluster cache stampede mitigation via probabilistic early expiration (Milestone 25) | 2 | $320.00 | **$640.00** |
| 26 | ENG-INF-026 | Zero-downtime schema migration verification for high-concurrency order ledger (Milestone 26) | 3 | $450.00 | **$1350.00** |
| 27 | ENG-INF-027 | Virtual thread concurrency soak probe under simulated 10k connections (Milestone 27) | 4 | $390.00 | **$1560.00** |
| 28 | ENG-INF-028 | Distributed trace propagation audit and OpenTelemetry exporter alignment (Milestone 28) | 1 | $260.00 | **$260.00** |
| 29 | ENG-INF-029 | Distributed consensus state-machine replica verification and Byzantine audit (Milestone 29) | 2 | $480.00 | **$960.00** |
| 30 | ENG-INF-030 | High-throughput reactive event ingestion pipeline tuning and partition testing (Milestone 30) | 3 | $350.00 | **$1050.00** |
| 31 | ENG-INF-031 | Native PostgreSQL read-replica connection pooling optimization via PgBouncer (Milestone 31) | 4 | $275.00 | **$1100.00** |


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| # | SKU | DESCRIPTION | QTY | RATE (USD) | AMOUNT (USD) |
|---|---|---|---|---|---|
| 32 | ENG-INF-032 | Redis cluster cache stampede mitigation via probabilistic early expiration (Milestone 32) | 1 | $320.00 | **$320.00** |
| 33 | ENG-INF-033 | Zero-downtime schema migration verification for high-concurrency order ledger (Milestone 33) | 2 | $450.00 | **$900.00** |
| 34 | ENG-INF-034 | Virtual thread concurrency soak probe under simulated 10k connections (Milestone 34) | 3 | $390.00 | **$1170.00** |
| 35 | ENG-INF-035 | Distributed trace propagation audit and OpenTelemetry exporter alignment (Milestone 35) | 4 | $260.00 | **$1040.00** |

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|---|---|
| Subtotal: | **$31790.00** |
| Sales Tax (8.5%): | **$2702.15** |
| Shipping and Handling: | **$125.00** |
| **Total Due:** | **$34617.15** |

### TERMS AND CONDITIONS

Payment is due within Net 30 Days of invoice issuance. Late balances are subject to a 1.5% monthly finance charge. All distributed computing consulting services, infrastructure deliverables, and architectural implementations are certified under Enterprise SLA terms. For inquiries regarding this statement, please contact accounting@apex.systems.


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